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Cancer Patients and Cancer Related Death Cases in Nigeria: A Case Study of Abuja and Ibadan Cancer Registries

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Abstract- The transformative shift in health patterns across sub-Saharan Africa has sparked a significant surge in the prevalence of non-communicable diseases, particularly cancers. Despite the global recognition of the pivotal role played by cancer registries in understanding the disease burden, conducting research, and shaping effective control measures, sub-Saharan Africa often grapples with the neglect of cancer registration due to resource constraints in the healthcare sector. The Nigerian National System of Cancer Registries (NSCR), a ground breaking initiative aimed at establishing a comprehensive, nation-wide approach to cancer registration in Nigeria. This system not only coordinates the efforts of existing cancer registries but also bolsters their capabilities, creates new registries, meticulously compiles and analyzes data, and freely disseminates this invaluable information to researchers and policymakers alike. This report not only shows the successful implementation of the NSCR but also sheds light on the formidable challenges faced during this transformative journey and the innovative solutions that propelled its success. Serving as a beacon for other low- and middle-income countries aspiring to enhance their cancer registration coverage, this narrative emphasizes the crucial components of training, mentoring, scientific and logistic support, and advocacy that are indispensable in sustaining effective cancer registration programs in similar settings. Dive into the narrative of overcoming challenges and triumphs, offering a roadmap for nations striving to fortify their stance against the growing cancer epidemic.

Index Terms- Nigeria; cancer; cancer incidence; cancer registration; population-based cancer registries.

I. INTRODUCTION

The world is contending with the rising impact of cancer, emerging as a significant contributor to both morbidity and mortality on a worldwide scale. The menace of cancer is particularly pronounced in sub-Saharan Africa (SSA), presenting a formidable challenge for public health. However, despite the pressing threat that cancer poses to the well-being of individuals in this region, a notable scarcity of comprehensive data on cancer incidence persists. This paper will reveal the data by showing the prevalence of cancer incidence specifically in two-cities of Nigeria. The foundation of these data relies upon the data collected from two distinctive population-based cancer registries (PBCR), integral components of the comprehensive Nigerian national cancer registry program. Through the lens of these registries, we embark on a study to unravel the complicated cases and dynamics of cancer incidence in Nigeria, shedding light on the unique challenges and opportunities that define the foundation of cancer epidemiology in the SSA region.

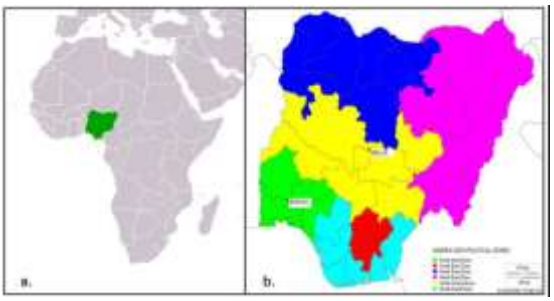


Figure 1: (a) The first image is the map of Africa showing Nigeria
(b) Second image is the Nigerian map showing Abuja and Ibadan

II. MATERIALS AND METHODS

We carefully examined data sourced from two distinctive population-based cancer registries in Nigeria—namely, the Ibadan Population-Based Cancer Registry (IBCR) and the Abuja Population-Based Cancer Registry (ABCR)—spanning a comprehensive two-year period from 2009 to 2010. Our

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analysis categorizes the data by registry, gender, and age groups, offering insights into the age-specific incidence rates of all invasive cancers. Additionally, we present age-standardized rates for the prevalent cancers, meticulously stratified by gender across both registries.

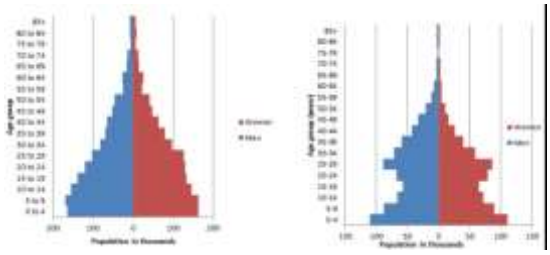


Figure 2: Comparative Population Pyramids of Ibadan and Abuja

III. RESULTS

The age-standardized incidence rates for all invasive cancers, as reported by the Ibadan Cancer Registry (IBCR), stood at 66.4 per 100,000 for men and 130.6 per 100,000 for women. Meanwhile, the Abuja Cancer Registry (ABCR) recorded rates of 58.3 per 100,000 for men and 138.6 per 100,000 for women. IBCR documented a total of 3,393 cancer cases, with 34% (1,155) occurring in males and 66% (2,238) in females. In the same timeframe, Abuja reported 1,128 invasive cancers, with 33.6% (389) in males and 66.4% (768) in females.

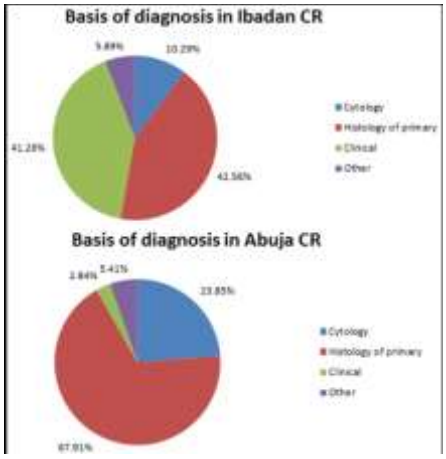


Figure 3: Most common basis of diagnosis in Ibadan and Abuja cancer registries

The mean age of cancer diagnosis for men in Ibadan and Abuja was 51.1 and 49.9 years, respectively, while for women, it was 49.1 in Ibadan and 45.4 in Abuja. Notably, breast and cervical cancers dominated among women, with prostate cancer being the most prevalent among men. The age-standardized incidence rate for breast cancer at IBCR was 52.0 per 100,000 and 64.6 per 100,000 at ABCR. For cervical cancer, IBCR reported rates of 36.0 per 100,000, compared to 30.3 per 100,000 at ABCR. Surprisingly, the observed differences in incidence rates of breast, cervical, and prostate cancer between Ibadan and Abuja did not reach statistical significance, adding complexity to the regional cancer population.

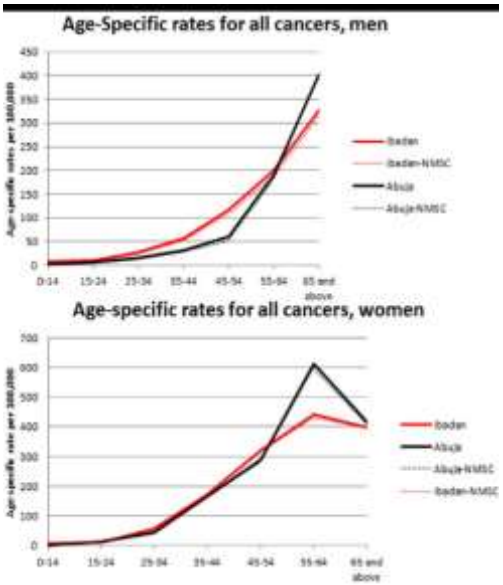


Figure 4: Showig the Age-Standardized Rates for All Cancers in Men and Women, With and Without Non-Melanoma Skin Cancers (NMSC), during the 2008–2010 Period in ABCR and IBCR.

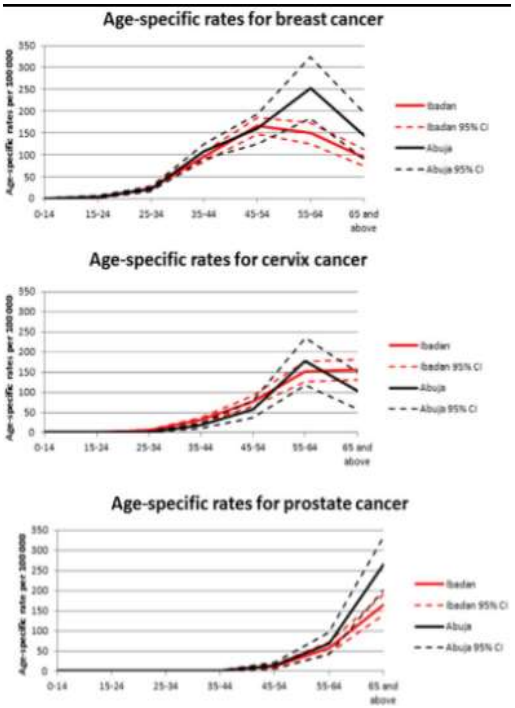


Figure 5: Showing Age-Specific Incidence Rates for Breast, Cervical, and Prostate Cancer

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IV. CONCLUSION

In this paper, we unveil the analysis of cancer incidence data from two population-based cancer registries in Nigeria reveals a significant surge in breast cancer cases in recent years. This study underscores the imperative for the establishment of top-notch regional cancer registries in Nigeria and other Sub-Saharan African countries.

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Conflict of Interest Statement
Conflicts of Interest
The authors declare no conflict of interest.

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